

ITM426, Long Quiz, 2026 Fall

Solution and Grading

- ITM 426 Engineering Mathematics 26F
- Duration: 90 minutes
- Weights: 10%
- 4 Questions

- Name: _____
- Student ID: _____
- E-mail: _____@seoultech.ac.kr

- Write legibly.
- Justification is necessary unless stated otherwise.
- Partial points are given only sparingly for the most problems because you are expected to 1) carry out proper sanity check and 2) correct your mistake by doing so.

#1. Define what it means for the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ to be *linearly independent*. [5pts]

Difficulty: Easy

Amount of work: 25%

Solution:

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are said to be *linearly independent* if none of the vectors can be expressed as a linear combination of the other $k - 1$ vectors.

#2. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

is a basis of a three-dimensional vector space. Prove that

$$\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3\}$$

is also a basis of the three-dimensional vector space. [5pts]

Difficulty: Easy

Amount of work: 25%

Solution:

Since

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

is a basis of the three-dimensional vector space, any arbitrary vector $\mathbf{x}$ in the space can be expressed as

$$\mathbf{x} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3,$$

where a_1, a_2, a_3 are real numbers. We want to show that the same vector $\mathbf{x}$ can be expressed as a linear combination of

$$\mathbf{v}_1 + \mathbf{v}_2, \quad \mathbf{v}_2 + \mathbf{v}_3, \quad \mathbf{v}_3.$$

Suppose

$$\mathbf{x} = b_1(\mathbf{v}_1 + \mathbf{v}_2) + b_2(\mathbf{v}_2 + \mathbf{v}_3) + b_3\mathbf{v}_3.$$

Expanding the right-hand side gives

$$\mathbf{x} = b_1\mathbf{v}_1 + (b_1 + b_2)\mathbf{v}_2 + (b_2 + b_3)\mathbf{v}_3.$$

Comparing the coefficients with

$$\mathbf{x} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3,$$

we obtain

$$b_1 = a_1,$$

$$b_1 + b_2 = a_2,$$

$$b_2 + b_3 = a_3.$$

Thus,

$$b_1 = a_1, \quad b_2 = a_2 - a_1, \quad b_3 = a_3 - a_2 + a_1.$$

Since a_1, a_2, a_3 are arbitrary real numbers, b_1, b_2, b_3 are also well-defined real numbers. Therefore, any vector $\mathbf{x}$ in the three-dimensional vector space can be expressed as a linear combination of

$$\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3\}.$$

Hence,

$$\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3\}$$

is also a basis of the three-dimensional vector space.

#3. Consider the following system of linear equations:

$$x + 2y = 8,$$

$$3x - y = 3.$$

- (a) Express the system in the matrix form $A\mathbf{x} = \mathbf{b}$. [1.5pts]
 (b) Calculate $|A|$. Based on the determinant, determine whether the system has a unique solution. [1.5pts]
 (c) Find the inverse matrix A^{-1} [1pts].
 (d) Using $\mathbf{x} = A^{-1}\mathbf{b}$, find the values of x and y [1pts].

Difficulty: Easy

Amount of work: 25%

Solution:

- (a) The system can be written as

$$\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}.$$

Therefore,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}.$$

- (b) The determinant of A is

$$|A| = (1)(-1) - (2)(3) = -7.$$

Since

$$|A| = -7 \neq 0,$$

the matrix A is non-singular. Therefore, the system has a unique solution. (c) For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the inverse matrix is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Thus,

$$A^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -2 \\ -3 & 1 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}.$$

- (d) Using

$$\mathbf{x} = A^{-1}\mathbf{b},$$

we have

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 8 \\ 3 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 8 + 6 \\ 24 - 3 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 14 \\ 21 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Hence,

$$\boxed{x = 2, \quad y = 3}.$$

#4. Consider the following system of linear equations, where p and q are real numbers:

$$x + 2y = 3,$$

$$px + 4y = q.$$

- (a) Express the system in the matrix form $A\mathbf{x} = \mathbf{b}$. [1pts]
 (b) Find the determinant $|A|$ in terms of p . Determine the values of p for which A is singular and non-singular. [1pts]
 (c) If A is non-singular, find A^{-1} and obtain the unique solution (x, y) in terms of p and q . [1pts]
 (d) Suppose that A is singular. Determine the condition on q under which the system has infinitely many solutions. [1pts]
 (e) Suppose that A is singular. Determine the condition on q under which the system has no solution. [1pts]

Difficulty: Easy

Amount of work: 25%

Solution:

- (a) The system can be written as

$$\begin{bmatrix} 1 & 2 \\ p & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ q \end{bmatrix}.$$

Therefore,

$$A = \begin{bmatrix} 1 & 2 \\ p & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ q \end{bmatrix}.$$

- (b) The determinant of A is $|A| = (1)(4) - (2)(p) = 4 - 2p = 2(2 - p)$. Therefore, $|A| \neq 0 \iff p \neq 2$. Thus, A is non-singular when $\boxed{p \neq 2}$, and singular when $\boxed{p = 2}$.
 (c) When $p \neq 2$, the inverse matrix exists and is given by

$$A^{-1} = \frac{1}{4 - 2p} \begin{bmatrix} 4 & -2 \\ -p & 1 \end{bmatrix}.$$

Hence,

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{4 - 2p} \begin{bmatrix} 4 & -2 \\ -p & 1 \end{bmatrix} \begin{bmatrix} 3 \\ q \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{4 - 2p} \begin{bmatrix} 12 - 2q \\ -3p + q \end{bmatrix}.$$

Thus,

$$\boxed{x = \frac{12 - 2q}{4 - 2p} = \frac{6 - q}{2 - p}}$$

and

$$\boxed{y = \frac{q - 3p}{4 - 2p}}.$$

(d) When A is singular, we have

$$p = 2.$$

The system then becomes

$$x + 2y = 3,$$

$$2x + 4y = q.$$

Multiplying the first equation by 2 gives

$$2x + 4y = 6.$$

Therefore, the two equations represent the same equation if

$$q = 6.$$

In this case, the system has infinitely many solutions. Hence,

$$\boxed{p = 2, \quad q = 6}$$

gives infinitely many solutions.

(e) Again, when $p = 2$, the left-hand side of the second equation is twice the left-hand side of the first equation. However, if

$$q \neq 6,$$

the two equations are inconsistent. Therefore,

$$\boxed{p = 2, \quad q \neq 6}$$

gives no solution.